

Data Challenge: Zero Hunger Lab

1 Overview

Recurrently over the years, the population of South Sudan is affected by floods. These floods have varying impacts on the population, including damage to homes, farmland and critical infrastructure, as well as an increase in disease outbreaks and population displacement¹². Where humanitarian organizations traditionally respond after the impact of floods occur to address the critical needs of the affected population, the focus of humanitarian organizations is shifting more and more to anticipatory aid (referred to as Anticipatory Action). Due to increased data availability and accuracy globally, we can predict the timing, location and severity of floods with increasing confidence. By taking actions based on these predictions in the critical window between when a flood is expected to be imminent and the actual onset of the flood, e.g., by vaccinating livestock before the flood takes place, the impact is reduced.

So far, Anticipatory Action has successfully been implemented to reduce the impact of flooding in several regions globally. However, there are regions where there is still limited availability and accuracy of flood-related data, making it challenging to design such an Anticipatory Action project. One of these regions is South Sudan, where Anticipatory Action has the potential to mitigate various types of impacts that the population experiences due to flooding. In collaboration with the Humanitarian Organization ZOA International, Tilburg University's Zero Hunger Lab wants to use data science to improve the understanding of riverine flooding in South Sudan. Capstone Data Challenge is conducted together with the Zero Hunger Lab (ZHL) of Tilburg University.

2 The Case

2.1 Context

Flooding is a disaster that affects the population of South Sudan on a yearly basis. However, the timing, severity, and location of the floods are different every year. Motivated by ZOA's goal of constructing an Anticipatory Action project to mitigate the impact of floods in South Sudan, Zero Hunger Lab (ZHL) is conducting research into the flood dynamics in South Sudan. Ideally, such an Anticipatory Action project decides when to initiate actions based on a (sufficiently accurate) impact-based forecast, which is able to provide a probabilistic estimate of the expected inundation on a very local level.

Before constructing such a forecast, it is crucial to first understand the flood dynamics in the area better. Therefore, a key question to ZOA and ZHL is: **what can publicly available data reveal about riverine flooding in South Sudan, particularly regarding relationships between hydrological or meteorological data and changes in the riverine discharge, inundation extent, and the flood impact?**

¹<https://www.who.int/publications/m/item/public-health-situation-analysis—south-sudan1>

²<https://reliefweb.int/report/south-sudan/south-sudan-floods-snapshot-17-september-2025>

2.2 Anticipatory Action

There are three main types of anticipatory actions that ZOA is currently considering:

1. cash transfers (most common across the sector, it lets people buy materials or livestock ahead of a flood, or avoids them having to sell livestock to afford food)
2. temporary flood defences (clearing drainage, sandbags, digging trenches)
3. early harvesting (highest stakes of the three, needs high confidence and only works in a specific window).

2.3 Investigating riverine flooding in South Sudan

The data challenge specifically focuses on increasing the understanding of riverine flooding in South Sudan. There are several types of data sources that are relevant when investigating these dynamics, which we categorize into hydrological, meteorological, and exposure data. With hydrological data we refer to data that reflects hydrological characteristics, such as the water level or discharge of the river. A critical dataset that we include is post-processed satellite data from NASA's MODIS and VIIRS missions, which contains flood records on a daily basis from 2002-2025 on roughly a spatial resolution of 250 meters, covering South Sudan entirely. This enables us to extract on a relatively local level where flooding has occurred in the past. Meteorological data captures different factors that can influence the volume of water in the river, such as rainfall. Exposure data describes the people, assets and livelihoods that are present in the (potentially inundated) areas, important to understand the potential impact of a flood in that given area.

These datasets and their relationship to discharge variability, inundation extent and flood impact can be explored in many different ways, and help to answer many different questions. The analyses that you choose to do should ultimately help inform the creation of an impact-based forecast for riverine flooding in a given area of South Sudan. The provided data spans South Sudan entirely, and some data sources are even provided on the full Nile Basin. While floods occur widely spread out over the country, certain areas are affected more frequently and more intensely. Furthermore, the impact on the population varies per region and over time: in some regions, the crop yields might only slightly deteriorate while in other regions, the population will need to evacuate. This affects both the analysis that is needed, as well as what data is available for this analysis.

2.4 Assignment

You are provided with several historical datasets that describe hydrological, meteorological and exposure data related to floods in South Sudan. These datasets differ in spatial and temporal resolution.

You are asked to make use of this data to increase understanding regarding discharge dynamics, flood occurrences, inundation extent and/or impact in South Sudan. The fundamental idea is to identify which (combination of) data features are important drivers of floods, and to explore and exploit these features such that this can ultimately be used for improving (impact-based) flood forecasting in the region. For this, you can explore several directions, such as:

- The relation between data and discharge dynamics;
- The relation between data and binary flood occurrences;
- The relation between data and inundation extent;

- The relation between data and impact on critical infrastructure;
- The relation between data and the impact on agricultural yield;
- The relation between data and impact on livelihoods;
- The potential of constructing impact-based or binary flood inundation forecasts.

You are, however, expected and encouraged to also explore other ideas and directions.

The expectation is that you develop techniques through which you can provide ZOA International and Zero Hunger Lab insights into riverine flooding dynamics in South Sudan and what the capabilities and limitations of these techniques are. Notice that you are not expected to solve the problem, but help Zero Hunger Lab explore possibilities and learn how to improve their work.

2.5 Resources

Below, we describe the publicly available data that we provide you. The data can be downloaded using the link and password provided on Canvas. Furthermore, we provide a GitHub repository, also linked on Canvas, with code for loading the data and conducting some initial exploration, as well as relevant background literature on the provided datasets.

Meteorological/hydrological data:

- **Flood inundation records:** post-processed satellite data from NASA’s MODIS and VIIRS missions, which contains data from 2000-2025 on a 250m grid for days on which a flood was identified [1]. This data distinguishes two different types of floods: recurring, meaning it happens on average at least once every three years over all entries in the dataset, and unusual, meaning that flooding happens less often on average at that location.
- **Historical rainfall data:** daily estimates of the total precipitation within the Nile Basin, extracted from the ERA5 reanalysis data [2]. This data is provided on a grid of $0.25^\circ \times 0.25^\circ$.
- **Historical runoff data:** daily estimates of the total runoff within the Nile Basin, extracted from the ERA5 reanalysis data [2]. This data is provided on a grid of $0.25^\circ \times 0.25^\circ$.
- **Historical evapotranspiration data:** daily estimates of the total evapotranspiration within the Nile Basin, extracted from the ERA5 reanalysis data [3]. This data is provided on a grid of $0.1^\circ \times 0.1^\circ$.
- **Historical lake levels:** the water level of upstream lakes can influence the water volume in rivers downstream. We provide the water levels of three lakes that are upstream the White Nile in South Sudan: Lake Kyoga, Victoria and Albert [4], [5]. These water levels are derived from satellite altimetry data, available roughly every 10 days between 2002-2025.
- **Historical discharge data:** we provide discharge data of 12 stations in the Nile Basin, derived from satellite data [4]. This data is available on a daily basis between 1998-2025, and includes one station in South Sudan.
- **Administrative boundaries data:** we include the most recent boundaries of the administrative levels of South Sudan [6].

Exposure data:

- **Crop and grazing area:** Per pixel (about 1/4 square kilometer), the percentage of land in that area which is crop-land and where cattle is grazing [7].
- **GDP:** the GDP of South Sudan on a yearly basis [8].
- **Cattle data:** the cattle population in South Sudan, on roughly a spatial resolution of 9.3 kilometres [9].
- **Population data:** the number of people living in South Sudan, on roughly a spatial resolution of 90 meters [10].
- **IPC data:** the number of people facing acute food insecurity in South Sudan between 2017-2025 on admin-2 level [11].
- **Infrastructure data:** we provide code how to download road networks from OpenStreetMap in Python, and a dataset containing the locations and types of health facilities in South Sudan [12].

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